

DIABETIC RETINOPATHY SCREENING REPORT

AI-Assisted Fundus Image Analysis — with Bootstrap 95% CI

Image: aptos_1e9224ccca95.png
Date: 2026-03-28 17:58:52
Model: EfficientNet-B3 (Binary + Multiclass Cascade)
Bootstrap: 2000 resamples, 95% confidence level

SCREENING RESULT (with 95% Bootstrap CI)

Binary Detection:

- Ensemble (5-fold+TTA): DR Detected
P(DR) = 0.9861 95% CI [0.9837, 0.9885]
- 2-Phase (single+TTA): DR Detected
P(DR) = 1.0000 95% CI [1.0000, 1.0000]

Severity Grading (Direct Multiclass Model):

- Predicted Grade: 4 — Proliferative DR
- Confidence: 42.9% 95% CI [0.3774, 0.4896]
- Severity: Sight-threatening

Per-Grade Probabilities (95% CI):

G0 (No DR)	): 0.0008	[0.0006, 0.0010]
G1 (Mild NPDR)	): 0.0540	[0.0355, 0.0722]
G2 (Moderate NPDR)	): 0.2565	[0.2127, 0.2991]
G3 (Severe NPDR)	): 0.2595	[0.2114, 0.3027]
G4 (Proliferative DR)	): 0.4292	[0.3774, 0.4896] ◀

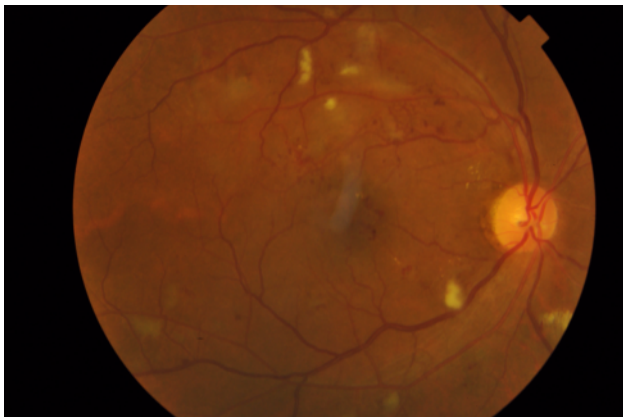
Clinical Recommendation:

- **URGENT referral — risk of vision loss**

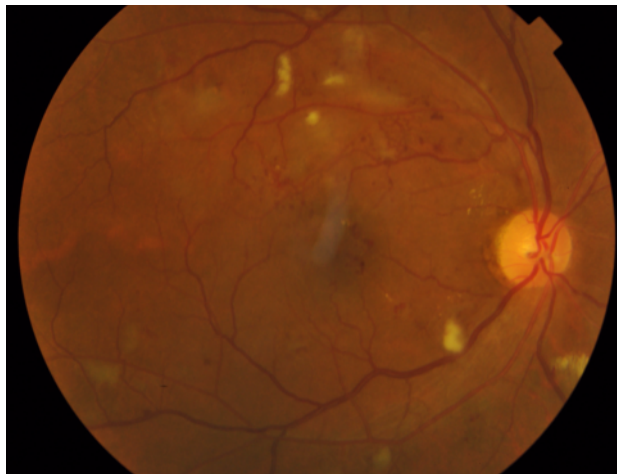
DISCLAIMER: This is an AI screening aid, not a diagnostic tool. All results must be confirmed by a qualified ophthalmologist.

Preprocessing Pipeline — aptos_1e9224ccca95.png

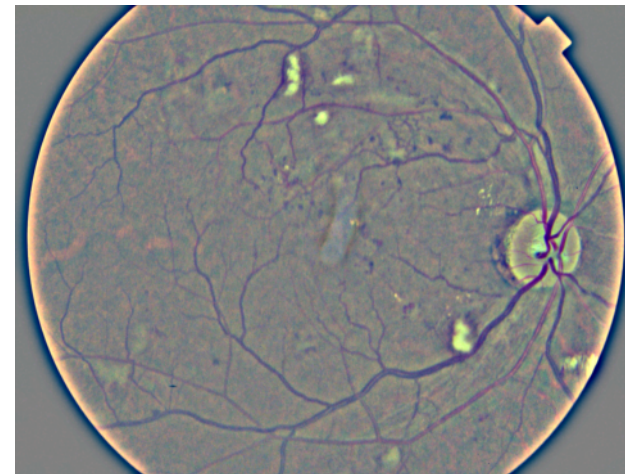
Step 1: Original



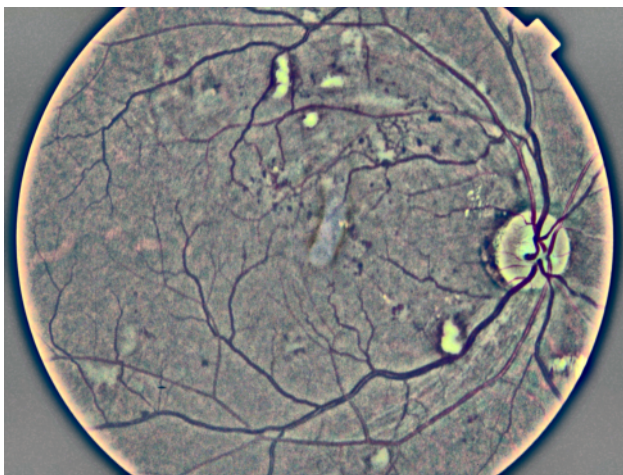
Step 2: Circle Crop



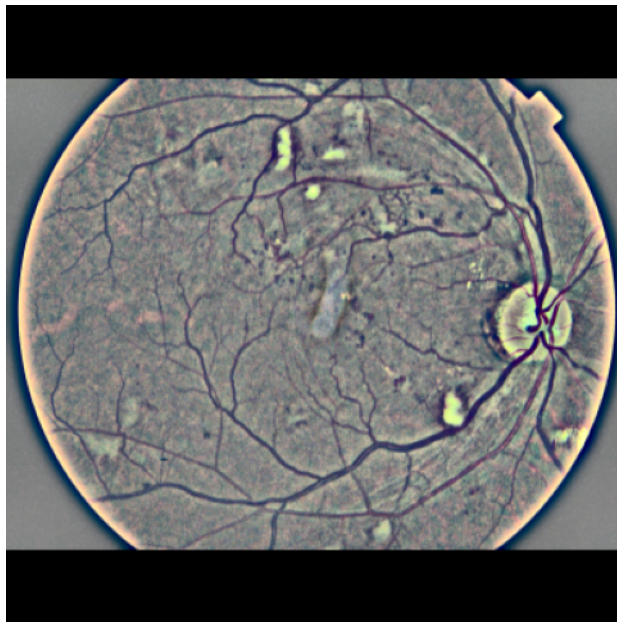
Step 3: Ben Graham



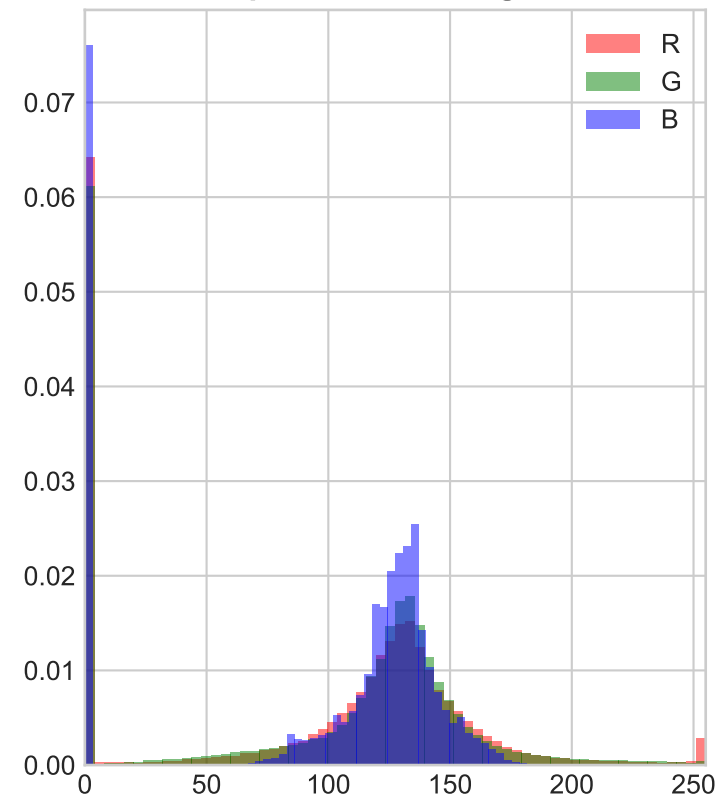
Step 4: CLAHE



Step 5: Final (384×384)

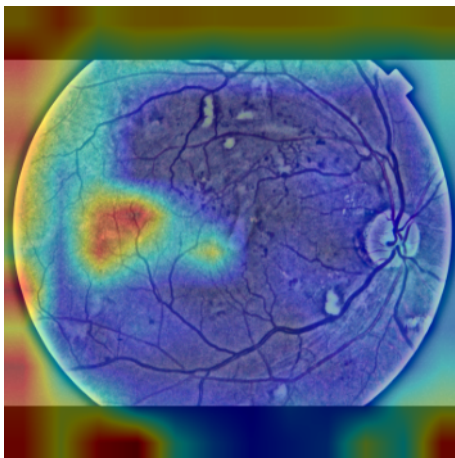


Preprocessed Histograms

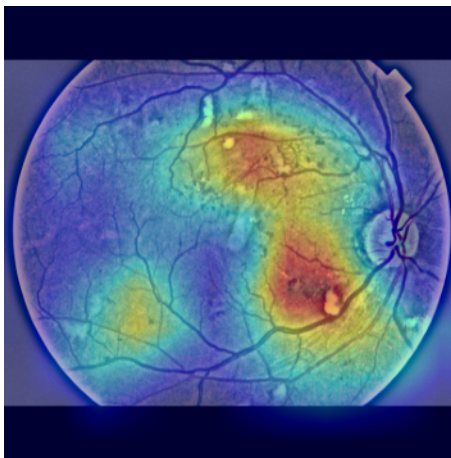


Explainability & Predictions with 95% CI — aptos_1e9224ccca95.png

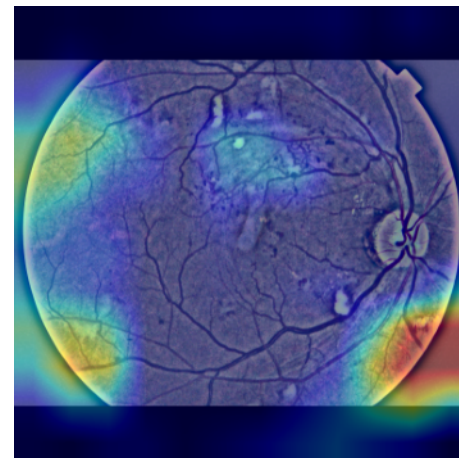
Ensemble GradCAM
P(DR)=0.9861 CI[0.9837,0.9885]



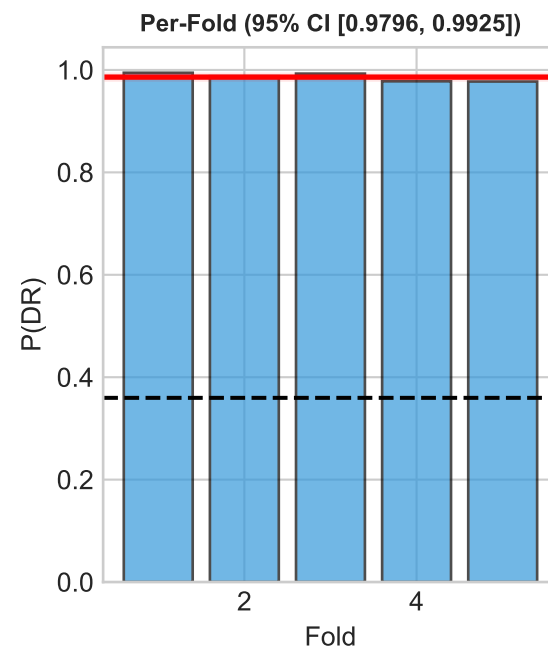
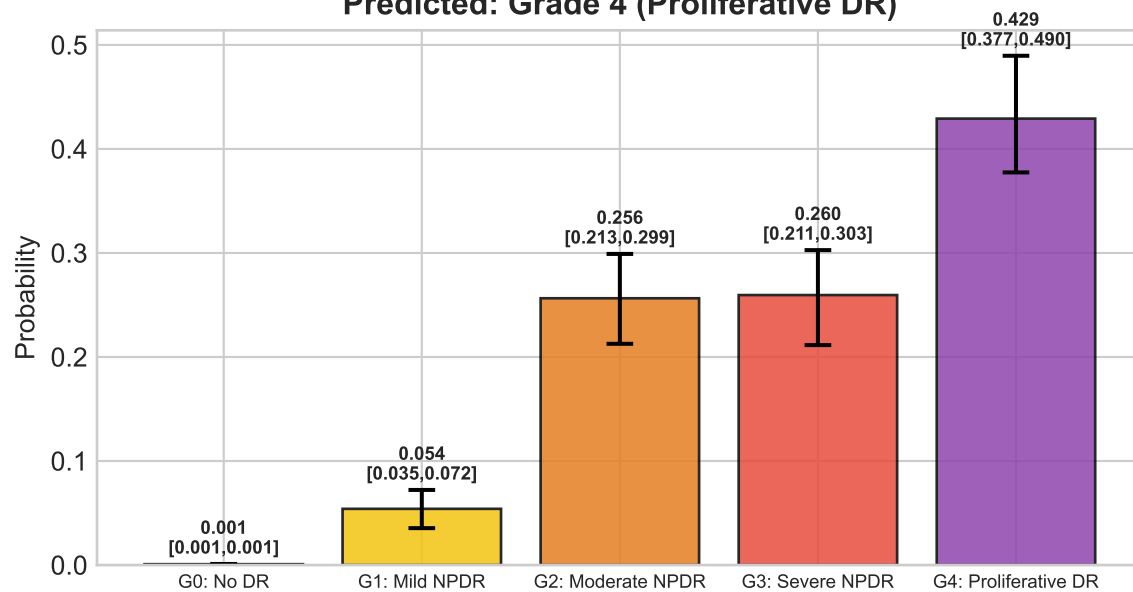
2-Phase GradCAM
P(DR)=1.0000 CI[1.0000,1.0000]



Multiclass GradCAM (G4)
Conf=42.92% CI[0.377,0.490]



5-Grade Probabilities with 95% CI (n_boot=2000)
Predicted: Grade 4 (Proliferative DR)



Model Validation Summary

MODEL VALIDATION METRICS

Dataset: APTOS 2019 (531 test images)

Bootstrap: 2000 resamples, 95% CI

Metric	Ensemble+TTA	2-Phase+TTA
AUC-ROC	0.9992	0.9992
Accuracy	0.9906	0.9887
Sensitivity	0.9962	0.9962
Specificity	0.9852	0.9815
F1-Score	0.9905	0.9886
Cohen Kappa	0.9812	0.9774

Multiclass Model (Direct 5-Class):

Best Validation QWK: 0.859

Architecture: EfficientNet-B3 + Dropout(0.4) + Linear(1536, 5)

Training: CrossEntropyLoss + label_smoothing=0.1

THIS IMAGE – PREDICTION SUMMARY

Ensemble P(DR): 0.986051 95% CI [0.983698, 0.988462]

2-Phase P(DR): 0.999999 95% CI [0.999999, 1.000000]

Multiclass Grade: 4 (Proliferative DR)

Confidence: 0.4292 95% CI [0.3774, 0.4896]

Per-Grade Probabilities (95% CI):

G0 No DR : 0.0008 [0.0006, 0.0010]

G1 Mild NPDR : 0.0540 [0.0355, 0.0722]

G2 Moderate NPDR : 0.2565 [0.2127, 0.2991]

G3 Severe NPDR : 0.2595 [0.2114, 0.3027]

G4 Proliferative DR : 0.4292 [0.3774, 0.4896]